Supplementary Appendix for Enforcement and Bargaining over Illicit Drug Wholesale Prices: Structural Evidence from a Transnational Asian Gang

Kaiwen Leong, Huailu Li, Marc Rysman, Christoph Walsh

A.1 Additional Figures and Tables

	Ice	Ecstasy	Ketamine	Erimin
Drug Class	A	A	B	C
Maximum prison years	Life	20	20	10
Maximum strokes of the cane	15	15	10	5
Threshold presumed trafficking*	25g	10g		
Mandatory Death Penalty	250g			

*Presumed trafficking may have a maximum sentence of life in prison.

Table A.1: Drug-trafficking laws in Singapore during our sample period.

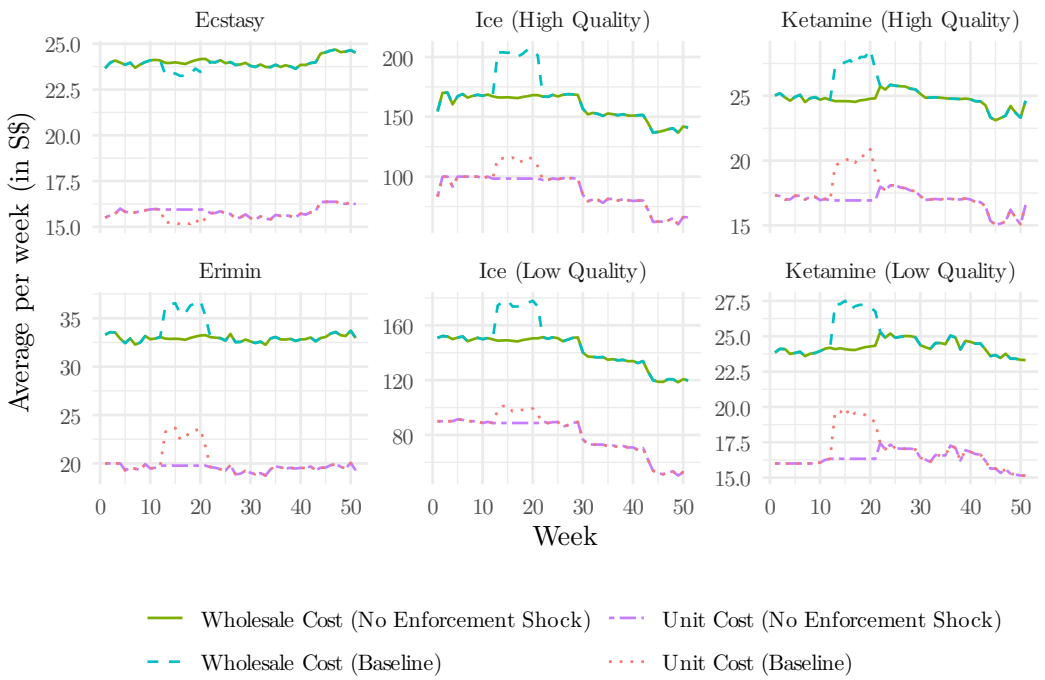


FIGURE A.1: Average weekly wholesale prices, no enforcement shock versus baseline.

A.2 Data Collection, Authentication and Replication

The unique dataset was originally collected by the first author (Leong). It was a by-product of his half-decade-long service as a social worker helping rehabilitate Singaporean ex-offenders by turning them into entrepreneurs.[1]

Specifically, the first author had volunteered with an ex-offender who volunteered his gang records as a token of gratitude upon learning about the first author's interest in studying social issues. This ex-offender, henceforth referred to as the informant, played a key role in a large transnational Asian drug gang in the 1990s. He learnt about the first author's interest in social issues after reading about the first author's books on the sex market (Leong et al., 2013b),

[1]Leong's social work has been documented in the Asian media (Singh, 2013; Palansamy, 2012). He was invited by neighbouring countries to help devise programs to rehabilitate ex-offenders.

gambling market (Leong et al., 2013a) and youths at risk (Leong and Leong, 2013). For obvious reasons, this informant remains anonymous. However, a redacted prison release letter proving this informant's status is available upon request subject to strict non-disclosure and privacy terms.

The rarity of obtaining such a dataset coupled with its sensitive nature warranted further investigation. Among these issues were information authentication, evaluation of potential risks and understanding the dataset's setting from an outsider's perspective. To verify the dataset's authenticity and to ensure it could be studied safely, the first author worked closely with the Institutional Review Board (IRB) at Nanyang Technological University (NTU) as detailed below. Regarding understanding the dataset's setting, the authors conducted interviews with more than 100 other ex-offenders about transnational gangs in Asia. These were completed securely through the first author's social network as well as in collaboration with drug rehabilitation centers in Singapore. The names of some of these organizations we contacted and proof of these conversations are available upon request, subject once again to strict non-disclosure and privacy terms, as the respondents have explicitly requested us not to reveal their information publicly. Further information about these interviews is provided in Section A.3.

The IRB's guidance and vetting were thorough and included the following tasks: (1) understand the legal implications of using the dataset, (2) share all information with the relevant appropriate authorities if not yet already done so by the informant, (3) take steps to protect all authors from any possible retaliation from former gang members, (4) verify the authenticity of the data with the appropriate authorities and experts, (5) document the payments to ex-drug offenders participating in the interviews, and (6) notify and obtain permission from the appropriate authorities to pursue this study.

We provided the required documentation to the satisfaction of the IRB. Proof that the abovementioned steps have been completed is available upon request, subject to strict non-disclosure and privacy terms.

We anticipate that there could be curious readers who wish to view the gang's original records. However, in accordance with the IRB's security proto-

cols, we were advised to relinquish possession of the original records for obvious safety reasons. We complied with this advice, but we can provide a redacted one-page photograph of the original book upon request. This is again subject to strict non-disclosure and privacy terms.

We also took steps to allow other researchers to replicate the results presented in this paper. In order to do so, the IRB required us to consult with the relevant authorities, ex-offenders and lawyers to ask for advice on developing a safety protocol to protect ourselves when dealing with possible replication requests. The resulting safety protocol requires anyone wishing to replicate the results to first obtain IRB approval and then to travel to Li or Walsh's universities to conduct the replication in person. As dictated by the safety protocol, we are strictly forbidden from discussing the data in private locations.

A.3 Interviews and Surveys with Ex-Drug Convicts

The ex-drug convicts that participated in the surveys were sourced from three main channels: ex-convicts that Leong had volunteered with directly; referrals from these same ex-convicts; and ex-convicts referred by the management of grassroots, government and religious organizations. Leong placed greater emphasis on seeking interviews with ex-convicts who were personally involved in the drug trade, specifically the buying and selling of drugs in the 1990s. The total number of respondents that fit these criteria was 105. Leong employed 13 ex-convicts as enumerators. Leong had personally volunteered with three of these ex-convicts, whereas the remainder were referred by the same three ex-convicts.

Preliminary interviews with five respondents revealed that they were particularly averse to paper documents and were wary about signing or writing on any documents. This is because although they had already carried out their prison sentences, they believed there were still risks involved for both themselves and their loved ones should their personal identity and history of drug abuse be leaked. Based on feedback from these respondents, survey collection should consist of face-to-face interviews in the form of a conversation.

Specifically, the interviewer would ask a question, the respondent would reply and the interviewer would record the answer on paper. Each interview was conducted either in English, Malay, Mandarin, or a Chinese dialect. We successfully tested this format on the same five respondents.

Leong ensured that all interviewers were trained to adhere to professional ethical standards. To ensure that respondents were willing to be interviewed, verbal consent was obtained prior to any interview. In addition, respondents were reminded of the interview's purpose, and reassured that their identities would be kept confidential. The first author would try to request bringing another person along with him during the interview so that he had a witness during the whole interview process. This way, the two interviewers could compare what they thought was said during the interview to ensure that the information collected was accurate.

Furthermore, if any of these ex-convicts were housed in drug rehabilitation centers, Leong would first seek permission from the management of that institution before contacting the respondents.[2] Some of these institutions also required the first author to send a hardcopy of the interview questions beforehand for vetting purposes while the others vetted the interview questions verbally before helping the first author to arrange interviews with the ex-drug convicts.

Below is a subset of the questions asked during the interviews:

- How many major drug delivery organizations were in operation during this time period?

  (1) 1 (2) 2 (3) 3 (4) more than 3 (5) Don't know

- Which types of drugs did drug gangs purchase from abroad? Multiple choices possible.

  (1) Heroin (2) Cannabis (3) Cocaine (4) Ice (5) Ketamine (6) Erimin (7) Ecstasy (8) Others, please specify (9) Don't know

[2] All leaders of organizations housing ex-drug offenders have instructed Leong not to state any identifiable information about the organization and inmates publicly.

- Which types of drugs did drug gangs purchase locally? Multiple choices possible.

  (1) Heroin (2) Cannabis (3) Cocaine (4) Ice (5) Ketamine (6) Erimin (7) Ecstasy (8) Others, please specify (9) Don't know

- What were the reasons that caused pushers to exit the industry? Multiple choices possible.

  (1) Religious beliefs (2) Family pleas (3) Legal employment opportunities (4) Other illegal employment opportunities (5) Planned to quit after sometime (6) Learned about the dangers of the industry

- Has your gang ever used violence on a pusher who wanted to exit the drug trade?

  (1) Yes (2) No (3) Don't know

- How much money did you make selling drugs per month?

  (1) S$1,000-S$5,000 (2) S$5,000-S$10,000 (3) S$10,000-S$15,000 (4) S$15,000-S$20,000 (5) More than S$20,000

- When you were arrested by the authorities for drug related offenses, could you afford bail?

  (1) Yes (2) No

- What are the different types of cost incurred when selling drugs? Multiple choices possible.

  (1) Car rental (2) Untraceable phone cards (3) Delivery costs (4) Bribes for law enforcement (5) Customer referral related expenses (6) Others

- Where did the majority of the drug sales take place?

  (1) Di San Qu – Geylang, Bukit Merah, Tanjong Pagar etc (2) Void decks of housing estates (3) Other Places (4) Don't know

A.4 End-User Prices

In this section we describe the sources we use to obtain the end-user prices that we use to estimate our model. We also discuss the consequences of any measurement error in end-user prices for our model estimates.

Ecstasy

The United Nations Office for Drug Control and Crime Prevention report that the average price of one dose of ecstasy in Singapore in 1998 was US$26.[3] Using the average closing exchange rate during our sample period, this is approximately S$43. Interviews with ex-drug offenders recalled prices that varied between S$35-60 per tablet, so S$43 falls close to the center of that range. Furthermore, 99% of trades in our data had a wholesale price of S$30 or less, so at S$43 pushers make a sizable gross margins over the wholesale price.

Erimin

Ex-drug offenders recalled that the price per slab (10 pills) of erimin costed between S$60-100 during our sample period. We take the midpoint of this range and use S$80 per slab, or S$8 per pill. Although several years since our sample period, a 2009 UNODC report gives a price of US$4 per pill in 2006 (S$6.36).[4] A separate newspaper article reports that prices ranged from S$6-8 per pill in 2006.[5] 99% of trades in our data have a wholesale price of S$60 or less per slab, so at S$80 pushers make large gross margins.

Ice

The United Nations Office for Drug Control and Crime Prevention report that the average price of one gram of ice in Singapore in 1999 was US$147.1, or

[3] https://www.unodc.org/pdf/report_1999-06-01_1.pdf
[4] http://drugslibrary.wordpress.stir.ac.uk/files/2017/03/2009_Patterns_and_Trends.pdf
[5] https://www.thestar.com.my/news/regional/2006/02/15/addicts-turn-to-erimin5

S$249.[6] End users recalled prices that range between S$150-380 per gram. A newspaper article reports that the price was between S$120-180 for half a gram.[7] The 25th and 75th percentiles of wholesale prices in the data are S$130 and S$180 per gram. In the data, the wholesale price for high-quality ice is on average 13.5% larger than for low-quality ice. Therefore we use S$240 for the end-user price for low-quality ice and S$280 for high-quality ice, which falls within the ranges given by the interviews and the reports, and has a similar difference in price to the average difference in wholesale prices for high- and low-quality ice.

We also note that our estimated price increases for high- and low-quality ice following the enforcement shock fall within the upper bound of the range recalled by the interview participants.

Ketamine

Ex-drug offenders report a range of S$45-85 per gram for the end-user price. An article from 2006 states that the price ranged from S$40-50.[8] The wholesale prices for high- versus low-quality ketamine are quite similar. The medians differ by only 4%. Therefore we use S$45 and S$50 per gram for low- and high-quality ketamine respectively.

End-User Prices and Identification

We note that if there is any measurement error in the end-user price it would directly translate into a corresponding shift in the pushers' average cost shocks, μ_j. For example, if the actual price per tablet of ecstasy was S$44 instead of S$43, the mean cost shock for ecstasy, μ_1, would be underestimated by 1. The pusher's payoff and demand, however, would not change as these two terms cancel out in our specification. Since our main counterfactual simulations do not rely on the magnitudes of the pushers' average cost shocks, μ_j, but only the

[6] https://www.unodc.org/pdf/report_2002-06-26_1/report_2002-06-26_1.pdf
[7] https://www.asiaone.com/health/rising-number-ice-users-singapore
[8] https://www.thestar.com.my/news/regional/2006/02/15/addicts-turn-to-erimin5

difference between the price and the shock, any measurement error in end-user prices will not affect the results of our main counterfactual simulations.

A.5 Gang Marginal Costs

In estimation, we use the reported unit costs in the data as the gang's actual marginal costs of trading. We do not include additional unobserved shocks to the gang's marginal costs because the variances of these shocks are not separately identified from the variances of the pushers' cost shocks.

The variance of the pusher's cost shock ξ_{ijt} is identified through selection into trading. A trade only occurs in a drug if the shock is low enough to satisfy $\xi_{ijt} < p_{jt} - c_{jt}$. If the gang's marginal cost also included a shock, only the sum of the pusher's shock ξ_{ijt} and the gang's shock would determine selection into trading and the two variances would not be separately identified from this selection condition.

If there was sufficient variation in wholesale prices conditional on trading and conditional on the pusher characteristics, then the variances of the gang's marginal cost shocks could be separately identified from the variances of the pushers' cost shocks. However, we have found that in practice there is not sufficient variation in our data to separately identify the both cost variances for each product. Therefore we assume that the gang's marginal cost is given by the unit costs in the data.

A.6 Obtaining Nash Bargaining Wholesale Prices

To calculate the wholesale prices $\boldsymbol{w}_{it}$ that satisfy the Nash bargaining first-order conditions for products where $p_{jt} > c_{jt} + \xi_{ijt}$, we find the fixed point of the function $\boldsymbol{\Omega}(\boldsymbol{w}_{it})$, whose jth element is:

$$\Omega_j(\boldsymbol{w}_{it}) = p_{jt} - \xi_{ijt} - \frac{(1-\beta_i)(p_{jt} - \xi_{ijt} + c_{jt} - 2w_{ijt})\, v_{it}(\boldsymbol{w}_{it})}{\beta_i \pi_{it}(\boldsymbol{w}_{it})}$$

We use the bisection method to find the fixed point of this function. The algorithm is as follows:

STEP 1 Set initial ($k = 0$) lower and upper bounds and initial guess:

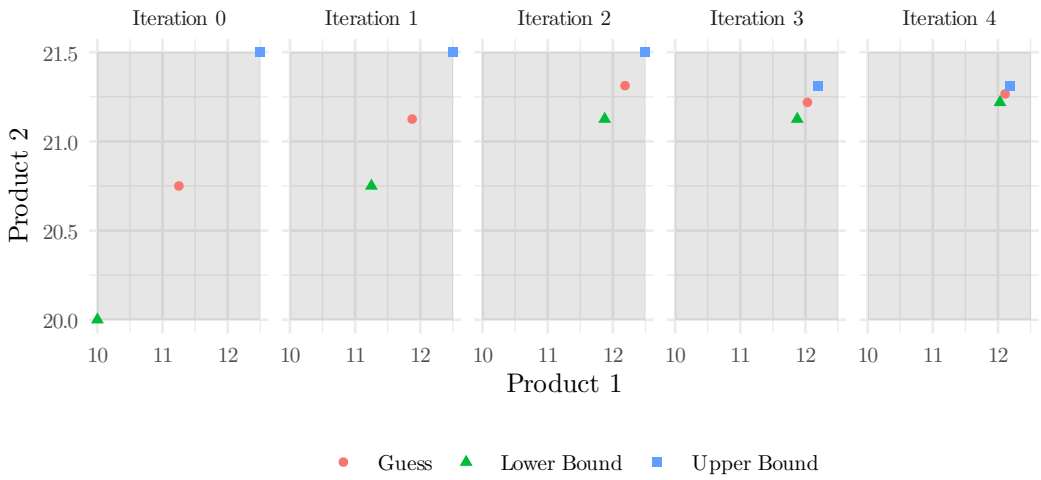
- Set the initial lower bound to $\underline{w}_{ijt}^0 = c_{jt} + \delta\left(p_{jt} - \xi_{ijt} - c_{jt}\right)$, where δ is a small constant (such as $10 \times e^{-5}$) such that $\Omega_j\left(\underline{\boldsymbol{w}}_{it}^0\right) < \underline{w}_{ijt}^0$ for all $i, j, t,$. The small constant is to avoid dividing by zero when $\pi_{it}\left(\boldsymbol{w}_{it}\right) = 0$.
- Set the initial upper bound to $\overline{w}_{ijt}^0 = \frac{p_{jt} - \xi_{ijt} + c_{jt}}{2}$, such that $\Omega_j\left(\overline{\boldsymbol{w}}_{it}^0\right) > \overline{w}_{ijt}^0$ for all $i, j, t,$.
- Set the initial guess $w_{ijt}^0 = \frac{1}{2}\left(\underline{w}_{ijt}^0 + \overline{w}_{ijt}^0\right)$.

STEP 2 For iteration $k > 0$, update the bounds and guess:

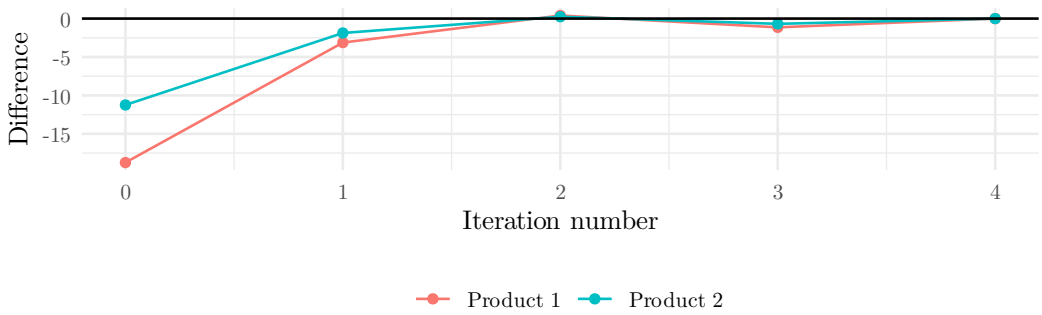
- If $\Omega_j\left(\boldsymbol{w}_{ijt}^k\right) < w_{ijt}^k$, set $\underline{w}_{ijt}^{k+1} = w_{ijt}^k$. Otherwise, set $\underline{w}_{ijt}^{k+1} = \underline{w}_{ijt}^k$.
- If $\Omega_j\left(\boldsymbol{w}_{ijt}^k\right) > w_{ijt}^k$, set $\overline{w}_{ijt}^{k+1} = w_{ijt}^k$. Otherwise, set $\overline{w}_{ijt}^{k+1} = \overline{w}_{ijt}^k$.
- Set $w_{ijt}^{k+1} = \frac{1}{2}\left(\underline{w}_{ijt}^{k+1} + \overline{w}_{ijt}^{k+1}\right)$.

STEP 3 Repeat step 2 incrementing k until $\|\boldsymbol{\Omega}\left(\boldsymbol{w}_{it}\right) - \boldsymbol{w}_{it}\|$ is below a pre-specified tolerance level.

An example with two products is shown in Figure A.2. This would occur in practice if there were two products where $p_{jt} > c_{jt} + \xi_{ijt}$ and $p_{jt} \leq c_{jt} + \xi_{ijt}$ for the remaining four products. The sum of marginal costs are $(c_{1t} + \xi_{i1t}, c_{2t} + \xi_{i2t}) = (10, 20)$ and the prices are $(p_{1t}, p_{2t}) = (15, 23)$. The initial lower bound $\underline{\boldsymbol{w}}_{it}^0$ is the green triangle and the initial upper bound $\overline{\boldsymbol{w}}_{it}^0$ is the blue square in Figure A.2a. The initial guess $\boldsymbol{w}_{it}^0$ is the red circle at the midpoint of these two points. In Figure A.2b, the difference $\Omega_j\left(\boldsymbol{w}_{it}^0\right) - w_{ijt}^0$ is negative for both products. Therefore the lower bound for iteration 1 is updated to $\underline{\boldsymbol{w}}_{it}^1 = \boldsymbol{w}_{it}^0$ for iteration 1 and the upper bound $\overline{\boldsymbol{w}}_{it}^1$ remains the same. The green triangle for iteration 1 moves to the position of the red circle from iteration 0. At $\boldsymbol{w}_{it}^1$, the difference $\Omega_j\left(\boldsymbol{w}_{it}^1\right) - w_{ijt}^1$ is again negative for both products, so only the lower bound is updated. In iteration 2, $\Omega_j\left(\boldsymbol{w}_{it}^2\right) - w_{ijt}^2$ is positive for both products, so $\overline{\boldsymbol{w}}_{it}^3$ is set equal to $\boldsymbol{w}_{it}^2$ and the blue square in iteration 3 moves to the position of the red circle in iteration 2. This process



(A) Iterating wholesale prices.



(B) Difference in $\Omega_j\left(\boldsymbol{w}_{it}^k\right) - w_{ijt}^k$ for $j = 1, 2$.

FIGURE A.2: Fixed point iteration example with two products.

continues until convergence. In practice, prices converge after 20-30 iterations. With $ns = 10{,}000$, each evaluation of our objective function requires solving for the optimal wholesale price vector using this fixed point approach over 30 million times because there are over 3,000 potential trades between the gang and the pushers.

A.7 Additional Counterfactual Experiments

A.7.1 Legalizing Ice

In this counterfactual we consider the effects of legalizing ice in our market. We focus on ice because the gang we study was the only gang in the country selling ice at the time, while there were several other gangs actively selling the other drugs. For ice, we observe the total quantity sold on the end-user market and we can use the enforcement shock to estimate the slope of the end-user demand curve. While no country has legalized ice to date, there has been a discussion of decriminalizing the drug in Australia (McGowan, 2019). Due to data limitations, this counterfactual requires a number of additional assumptions, but we will argue that we estimate a lower bound on the tax revenue that could be raised, and that the actual tax revenue is likely to be much larger.

We assume that the gang becomes a centralized importer and the pushers become licensed wholesalers. This is similar to the case of prostitution in Singapore. When prostitution was legalized, a fixed number of permits were issued to individuals already active in the market (Li et al., 2018; IRB Law, 2020). We assume the centralized importer's marginal costs are the same as the gang's, and since the enforcement shock would not occur for ice under legalization, we use the same marginal costs as described in the no enforcement shock counterfactual. We assume each pusher's demand is unchanged, but since the penalties from being caught are removed we argue that pushers would actually sell more ice under legalization. We further assume that the end-user demand function is unchanged following legalization. However, given the results in Jacobi and Sovinsky (2016), we would expect end-user demand to increase following legalization as access would increase and the social stigma of ice consumption may fall. We consider a tax rate of 7%, which is equal to the sales tax rate in Singapore and is in the lower end of the state taxes on marijuana in the US and Canada.[9]

[9] In the US, excise taxes on marijuana vary between 7% and 37% and in Canada between 3.9% and 19.3%.

In our data, we observe the total equilibrium quantity during the no-shock period and the enforcement shock period. We also observe the equilibrium price during the no-shock period and have estimated the rationalized price change during the enforcement shock period with our model. Since the enforcement shock is a cost shock which shifts the supply curve, we observe two points on the end-user market demand curve. If we make a linear approximation of the end-user demand curve, the enforcement shock allows us to estimate the slopes of the end-user demand curve for high- and low-quality ice. This approximation enables us to predict end-user demand after the imposition of a tax.[10] We also highlight that in our model each individual pusher faces a perfectly elastic demand curve since the end-user market is competitive. However, the market as a whole may face a less elastic demand curve. We find the demand elasticities for high- and low-quality ice to be -0.45 and -0.50, respectively.

To calculate how the burden of the tax is shared between end users and pushers, we use the following procedure. Given a trial share of the tax burden, we calculate total end-user demand using our linear approximation and total pusher supply by simulating trades according to our estimated bargaining model. We then find the vector of new equilibrium end-user prices such that total demand equals total supply in each product.

We find that end users bear 72.5% of the tax on high-quality ice and 70.1% of the tax on low-quality ice. This is in line with low-quality ice being slightly more elastic than high quality. The tax reduces the total quantity sold on the market by 2.64% and 2.78% for high- and low-quality ice, respectively. For some intuition about how the pushers' relative bargaining power with the gang affects the tax incidence between pushers and end users, we also calculate the incidence shares if we shift all the pushers' bargaining parameters to just above zero and just below one. We find that if pushers have more bargaining power over the gang, that the end users bear even more of the tax. The opposite occurs if the pushers have less bargaining power.

[10] To calculate these slopes, we only use weeks 10-48 in the data, where the number of active pushers is close to the median value.

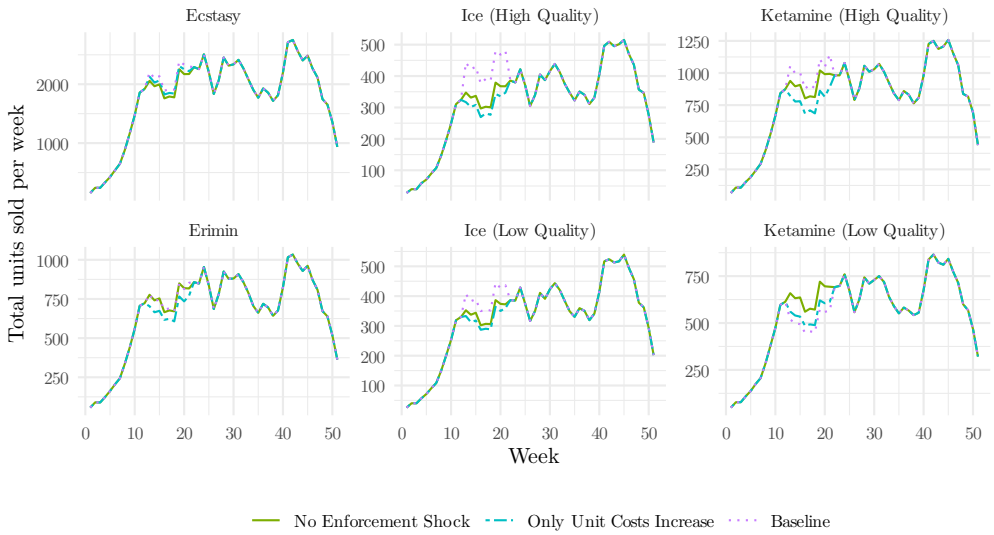
The estimated lower bound on the annualized tax revenue is S$738,782 (approximately US$760,000 in 2019 dollars), where 53.3% of this is from high-quality ice. Singapore's population was just under 4 million during our sample period. Therefore the tax intake from legalization may be very small and it is likely that the accompanying administrative costs would outweigh it.

A.7.2 Enforcement Shock Decomposition

We decompose the effect of the enforcement shock by simulating trades where only the gang's unit costs increased in the period following the enforcement shock and end-user prices did not adjust. Figure A.3 shows the total weekly quantities sold and average wholesale prices for (i) the counterfactual scenario of no enforcement shock as presented in the main text; (ii) the counterfactual scenario where only the gang's unit costs increase, the end-user prices do not adjust and pusher demand parameters remain the same; and (iii) the baseline scenario where the enforcement shock occurred. In Figure A.3a, we can see that for ice, which saw an increase in the quantity sold as a result of the shock, actually experiences a decrease in the quantity sold if there is no end-user price adjustment. This is intuitive as pushers have a higher cost per unit yet receive the same revenue per unit on the end-user market. This can be seen in Figure A.3b. When the enforcement shock occurs and end-user prices do not adjust, the increase in wholesale prices is slightly less than the increase in the gang's unit costs. This is because the gang has more bargaining power than most pushers and most of the cost increase is passed on to the pushers. However, when the end-user prices adjust, the gang passes more of the increase in unit costs onto the pushers as they are compensated by the higher end-user prices. For other drugs, there is very little difference because there was not a large shift in end-user prices following the shock.

A.7.3 Alternative Ways of Targeting Pushers

In the targeting pusher counterfactual in the main text, we assumed the authorities arrested 50 randomly-chosen pushers in one particular week. We



(A) Total sales of each drug in each week.

(B) Average Weekly Wholesale Prices.

FIGURE A.3: Enforcement shock with and without end-user price adjustments.

	(1) 50 random pushers in week 26	(2) 51 random pushers once per week	(3) 49 ex-convict pushers in weeks 26-29	(4) 49 pushers with club connections in weeks 26-40
Ecstasy	0.897	0.897	0.897	0.858
Erimin	0.896	0.896	0.896	0.856
Ice (High Quality)	0.904	0.898	0.903	0.859
Ice (Low Quality)	0.902	0.898	0.902	0.855
Ketamine (High Quality)	0.902	0.897	0.901	0.859
Ketamine (Low Quality)	0.894	0.896	0.894	0.851

Ratio of total quantity sold compared to total quantity sold in the baseline case.

Table A.2: Effect of alternative ways of targeting pushers on quantity sold.

can compare such a strategy to continuously arresting 1 pusher per week over the year, or arresting pushers with particular characteristics. We present the results from these simulations in Table A.2.

In column 1 we present the results from the main text to be able to compare the results. When we arrest 50 pushers in week 26, the total quantity sold in the year is approximately 90% of the total quantity sold under the baseline case for each drug. If we instead arrest 1 pusher per week over the 51 weeks in our sample period, we see a similar decrease of just under 90% for each drug. In the counterfactual in column 1, we assume that the sudden spike in arrests was unexpected and we can use the estimated pusher demand parameters to simulate trades. However, if pushers expect the arrests to increase, which would happen if additional arrests happened each week, then we might expect the pusher demand parameters to change. The gang may also change its recruiting strategy to offset the increase in arrests, although they would likely not completely offset it in order to avoid the attention of the authorities. Therefore continuously arresting pushers may actually have a larger decrease than what we find if the remaining pushers change their demand in response to the additional arrests.

In columns 3 and 4 we consider the effect of targeting specific types of pushers instead of targeting pushers randomly. In column 3 we consider the effect of arresting ex-convict pushers. The authorities would already have identifying information on these pushers because of their criminal records and they may be easier to target. Since there were fewer than 50 active ex-convict pushers in week 26, we assume the authorities arrest all the 49 ex-convict pushers that were active in weeks 26-29. The effect of such a strategy is very similar to arresting pushers at random with the total quantity sold approximately 90% of the baseline. In column 4 we suppose the authorities arrest the 49 pushers that were active in weeks 26-40 that had connections with night clubs and discotheques. Arresting these pushers has a larger effect because these pushers buy larger quantities on average. The quantity sold falls to approximately 85% of the baseline compared to 90%.

A.8 Transnational Gangs and Street Gangs

There are two types of drug-selling gangs: transnational and local street gangs. According to market insiders, there are three main differences between these two types of gangs in Asia. First, the total number of members in a transnational gang ranges from the thousands to the tens of thousands, while a street gang has members ranging from the tens up to the low thousands. Second, transnational gangs have more resources and similarly efficient organizational structures. They have effective strategies shaped by their experience in dealing with authorities in the various similar countries that they operate in. On the other hand, there is no standard organization structure for a street gang. It fluctuates from gang to gang. Third, to avoid physical clashes they cannot win, street gangs usually operate in areas that the transnational gangs do not consider profitable.

Given the lack of data on transnational drug-selling gangs, it is difficult to assess the external validity of our setting, but the evidence we have collected suggest that the gang we study is similar to those in other Asian countries. According to market insiders with whom we have spoken to, the basic demo-

graphics of gang members are similar across all gangs. Most gang members join this trade because they want to get rich quickly. According to an ex-drug gang member referred to us by Christian Counseling Service, "I ended up in jail for what I did because I wanted to be a billionaire overnight."[11] Some examples of common characteristics of gang members include high debt levels, an affinity for gambling, and drug addiction.

Gangs may differ in the way they regulate gang members. In particular, the most significant point of variation is how gangs deal with members who are also drug addicts. There exists gangs that punish their members for abusing drugs and those that do not. Both these types of gangs are prevalent in the drug-selling gang population. However, drug abusers may continue to exist nonetheless in both types of gangs. For gang bosses that choose to punish drug users, first-time offenders are usually given a severe physical punishment and warned not to commit the same act. Second-time offenders are usually expelled from the gang.[12] Only some drug addicts are discovered within gangs because some gang members who are drug addicts may have also developed effective ways to conceal their addiction from gang leaders. According to a law enforcement officer in Asia who spoke to us on the condition of anonymity, "it does not make sense to claim that any gang does not contain specific types of people such as drug addicts."[13]

References

IRB Law (2020): "Prostitution in Singapore," https://irblaw.com.sg/learning-centre/prostitution-in-singapore/.

Jacobi, L. and M. Sovinsky (2016): "Marijuana on Main Street? Estimat-

[11]Christian Counseling Service is a registered charity in Singapore that works with Singapore Prison Service to rehabilitate ex-drug offenders.

[12]According to a former drug-selling gang member, "my gang does not allow people to take drugs but there were people who got punished for doing it anyway."

[13]We spoke an ex-drug law enforcement officer in Malaysia. He was uncomfortable endorsing the statement that he has ever encountered drug-selling gangs that consisted only of gang members who do not abuse drugs.

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